COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-APP-009: Continuous Integration Servers (Custom Applications)

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Title	Continuous Integration Servers (Custom Applications)
Executive Sponsor	Enterprise Applications, Director of Enterprise Applications
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, including employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
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1. Purpose

OIT is statutorily obligated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). This document is submitted as the standard for the technology, specifications, methodologies, and practices to provide Continuous Integration and to satisfy stakeholder needs surrounding this service.

This document defines the standard methods that Enterprise Applications shall use to support delivery of a Continuous Integration Server (build automation, artifact management, deployment process management, role-based access, and state authentication constructs) through the adoption, use, and maintenance of an in-house Continuous Integration Server.

This technical standard is intended to be used as a comparator against proposed future Continuous Integration Servers. A rational determination of a specific server will enable more efficient support services and security, provide additional stability across development teams, improve consistency, and enable automation of common tasks.

2. Scope

These standards apply across OIT and agencies supported by OIT. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient. This is applicable to new solutions requested and major changes to existing solutions, developed by OIT staff as

well as vendors developing solutions for which OIT will provide day-forward maintenance and support.

This document is intended for Enterprise Applications, but applies to any OIT unit providing these services to customers.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

Enterprise Applications - Director of Enterprise Applications

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Continuous Integration Server Qualification

The CI process must be provided by a product identified in the current OIT Certified Standard Product for Continuous Integration Servers and provide the following required capabilities:

- Ability to manage scheduled automated builds.
- Ability to manage automated builds based on events such as an update to a code branch.
- Easy installation and numerous plugins to support CI pipeline tasks.
- Custom plugin creation abilities.
- Build snapshots.
- Manage (create, store, export, and import) continuous integration pipeline configurations.
- Scheduled and automated test execution.
- Ability to create custom scripts (e.g., database migration, sanity tests).
- Securely and automatically pull from code repositories.
- Support scripting to extend the pipeline (e.g., Bash, shell, Perl scripts).
- Built-in, configurable email notifications for events.
- Store states of deployment servers and allow rollbacks.
- Support backups, restores, and disaster recovery.

- Enable a collaborative environment where multiple developers can simultaneously contribute to projects and quickly verify results.
- Improved code maintenance with all code centralized in a single location and built by a single tool.
- Improve software quality.
- Provide visibility and insight into the health of development efforts in real time.
- Increase awareness and simplify decision making for when to deploy or continue testing.

Security Verifiable

Any Continuous Integration Server considered for use in development, deployments, or any aspect supported by OIT and/or its vendors is required to be compatible and compliant with security scanning and assurance tools. All Continuous Integration Servers are subject to vulnerability scanning, assessment, and mitigation prior to approval for deployment to a production environment.

Portability

Continuous Integration Servers shall be portable due to their collaborative nature and may be hosted across many platforms, infrastructures, and architectures. Interaction with the server should be system-agnostic.

Security Frameworks

Continuous Integration Servers must be configurable to prevent common security vulnerabilities such as unauthenticated access, malicious builds, malicious build agents, and malicious users. State applications are to comply with the level of security specified in OIT's Information Security Policies.

Accelerators

Continuous Integration Servers shall offer accelerators such as plugins for build automation, scheduled job automation features, pipeline creation tools, multi-pipeline capabilities, built-in support for Version Control, health monitoring tools, and other features for rapid development and delivery of build automation.

Sustainability

Products and services may be considered sustainable when rationally accounted for from both a budgetary and staff support perspective, including licensing/maintenance planning and ongoing lifecycle, security, patching, and support/KLO planning.

Authentication

Continuous Integration Servers shall be able to support authentication with Google and Microsoft platforms.

5. Variance and Exception Process

Exceptions to any portion of this standard may be requested via the OIT Secure Configuration Exception Request Form, as posted on OIT Plaza.

6. References

- OIT and Information Security Policies (see OIT Plaza - Policies): This document establishes operational standards for OIT. All staff following these standards shall do so in compliance with all other OIT policies as published. In the event of any perceived conflict, OIT Policies shall prevail.

7. Definitions

- Continuous Integration (CI): The practice of merging working developer copies to a shared mainline on frequent intervals.
- Continuous Integration (CI) Server: Provides the functions to support CI technical processes.
- Source Code Control: A version control system designed to track changes in source code and other text files during software development.